QP CODE: **24900267**

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**
**FIRST SEMESTER MGU-UGP (HONOURS) REGULAR**
**EXAMINATION NOVEMBER 2024**

First Semester -Discipline Specific Core Course

**MG1DSCZGY100, MG1DSCZIM100 -INTRODUCTION TO ZOOLOGY**

*(2024 Admission onwards)*

**Duration: 1.5 Hours** **Maximum Marks: 50**

## Part A

*Fill in the blanks*

Answer **all** questions, each question carries **1** mark

1 Male weaver birds build multiple nests to attract **females** as part of their courtship behavior. [U] [1]

2 Centre for Cell and Molecular Biology is located at **Hyderabad** [K] [3]

3 **Mulberry tea / any mulberry product** is a by-product of mulberry cultivation, said to be effective in lowering blood pressure. [U] [4]

4 In mammals the primary care giving roles is given by **mothers / female parent** [K] [2]

5 Egg yolk is used as an **emulsifier / binder / ingredient** in mayonnaise. [U] [4]

6 When different individuals specialized to perform specific tasks that benefit the group is **division of labour / Any answer relate to social organization** [K] [1]

7 **Epigeic / an example of** earthworms are well suited for Vermiculture. [K] [3]

8 The winged reproductive adults that leave a termite colony to form a new colony is called as **alates** [K] [1]

9 Faking is used in the context of mating for **increasing reproductive success / survival (Give full mark if attempted)** [K] [2]

10 **Bioluminescence** is the production and emission of light by living organisms. [K] [1]

**( 10 x 1= 10 Marks)**

## Part B

*Short Answer Type Questions*

Answer **any 10** questions**,** each questions carries **2 mark**

**(Any 2 Points)**

**11 Discuss the ecological impacts of Noctiluca blooms**. [U] [1]

Can release ammonia, or deplete oxygen/ result in hypoxia which may be harmful to marine life.

**12 Describe the feeding behaviour of adult velvet spiders**. [U] [2]

Nocturnal predators, some species may also scavenge.

**13 How can researchers predict the health of an ecosystem by analysing the diversity of earthworms in the soil?** [U] [3]
More earthworms- soil less polluted, nutrient rich, supports more plants & animals. Healthy ecosystem

**14 What is Vermiwash?** [K] [4]
The liquid fertilizer- from vermicompost

**15 Explain structural colouration with an example**. [U] [2]
Colours produced by physical structures of the surface eg. iridescent feathers of a peacock or butterfly wing.

**16 How group living reduces predation risk?** [K] [1]
Collective vigilance, increased group defence/offense, enhanced detection of predators, dilution effect or confusing predators.

**17 Name any four value added products from poultry.** [U] [4]
Egg products, cosmetics, special eggs, fertilizers, pharmaceuticals, animal feed, processed meat products.

**18 Name any two fishery research institutes in India**. [A] [3]
CMFRI- Central Marine Fisheries Research Institute, CIFT - Central Institute of Fisheries Technology, CIBA- Central Institute of Brackish water Aquaculture, CIFE -Central Institute of Fisheries Education, KUFOS- Kerala University of Fisheries and Ocean Sciences.

**19 Can pearl be used as an ecological indicator? Substantiate.** [U] [1]
Yes. Sensitive to environmental changes, understand minerology shifts under changing temperatures and environment over season and years. **Yes – 1 marks, Substantiate with one point – 1 mark**

**20 Why is basking significant in the parental care of African pythons?** [U] [2]
They cannot increase metabolism, depends solely on basking to increase body temperature needed for incubation of eggs

**21 Describe the duties of a Wildlife Biologist/Researcher.** [U] [3]
Conduct field research on wildlife species, habitats and ecosystems. Collect and analyse data to contribute to conservation efforts and management plans. Collaborate with local communities and stakeholders to promote biodiversity conservation.

**22 What entrepreneurial possibilities exist in pet boarding and daycare services for zoologists?** [U] [4]

Establish boarding facilities or daycare centers -safe, engaging environments -specialized care for pets with specific needs, elderly pets / medical conditions. Training, socialization programs, pet wellness packages.

**( 10 x 2 = 20 Marks)**

## Part C

*Short Essay Type Questions*

Answer **any 5** questions**,** each question carries **4 marks**

**23 Discuss the process of comb construction in bees**. [U] [1]

Site selection and cleaning by worker bees. Role of propolis. Geometrical principles. Production of beeswax for the comb. Process of adding new combs. (**Brief explanation of any 4 points)**

**24 Beehive is the pinnacle of biological architecture'. Substantiate the statement**. [U] [1]

Site selection and orientation of hive. Mention materials used for construction. Significance of hexagonal shape. Internal layout of hive - brood area, brood nest, pollen storage, honey storage.
**( Overall description of architecture of beehive can be considered**)

**25 Describe the advantages of mimicry**. [U] [2]

Avoidance of Predation, Foraging and Hunting efficiency, Reproductive Success, Mutual Benefit

**26 Discuss the role of mimicry in shaping biodiversity. Liberal Valuation** [U] [2]

Natural Selection and Genetic Variation.
(Brief description **of these points + Any other relevant** points pertain to the question).

**27 What are the main by-products of dairy farming that contribute to career opportunities?** [U] [3]

Production and marketing of cream, skimmed milk, buttermilk, casein, whey, ghee residues.
(Any 4 by-products with brief description)

**28 In what ways can dairy farmers diversify their income through value-added products?** [U] [4]

Wide range of value added products, fetching additional income, year round demand for products, sustainable income **(Name of any 4 by-products also should be mentioned**)

**29 Briefly describe the products and by products of Aquaculture.** [U] [4]

Products- food- humans, other organism
Byproducts of aquaculture- Fish oil, fish protein concentrate, fish albumin, fish flour, gelatin, fish manure, fish skin, isinglass, squalene, shark fin rays, fish insulin, chitin/ chitosan, glucosamine, algal byproducts.
(Any **4** products / by-products with brief explanation or **Name** of **8** products / by-products)

**( 5 x 4 =20 Marks)**

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